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RESEARCH ARTICLE

Stabilizing Efficient Perovskite/Silicon Tandem Solar Cells With Amorphous and Asymmetric Hole-Selective Contacts

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ABSTRACT

Perovskite/silicon tandem solar cells (TSCs) hold great promise for next-generation photovoltaics, yet uneven coverage and thermal instability of self-assembled hole-selective contacts hinder their commercialization. Here, we introduce ZF6, an amorphous molecular contact with an asymmetric π-extended structure, featuring a dimethylindenocarbazole core and carbazole substitution. ZF6 exhibits rapid and robust adsorption on transparent conductive substrates, achieving uniform coverage and exceptional thermal stability. Its optimal energy level alignment and enhanced carrier extraction efficiency are compatible with various bandgap perovskites. ZF6-based double-junction perovskite/silicon TSCs achieve a reverse-scan power conversion efficiency (PCE) of 34.02% (certified 32.81% at NREL) with improved operational stability, retaining over 97% and 87% of initial efficiency after 1000 h of maximum power point tracking at 45°C and 85°C, respectively. We also fabricate triple-junction perovskite/perovskite/silicon TSCs, achieving a certified steady-state PCE of 30.42%, which represents the highest-performing triple-junction perovskite-based TSCs reported to date. This work establishes a novel design strategy for robust molecular contacts, advancing efficient and durable perovskite-based tandem photovoltaics.

1 | Introduction

Perovskite/silicon tandem solar cells (TSCs) hold immense promise for surpassing the Shockley–Queisser theoretical efficiency limit of single-junction solar cells while maintaining low production costs [1, 2]. The theoretical power conversion efficiencies (PCEs) of double-junction and triple-junction perovskite/silicon TSCs can reach ~43% and ~50%, respectively [3–5]. However, interface uniformity and stability remain critical barriers to their commercialization [6, 7]. Self-assembled molecules (SAMs) are widely regarded as one of the most promising hole-selective layers (HSLs) due to their tunable chemical structures, molecular-scale thickness, ultra-low parasitic absorption, and defect passivation capabilities [8]. SAMs also form

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conformal layers on textured silicon substrates, making them ideal for perovskite/silicon TSCs [9]. Despite these advantages, conventional SAMs suffer from insufficient interface coverage [6], poor thermal stability [10], and suboptimal energy level alignment with wide-bandgap (WBG) perovskites [11], necessitating advanced molecular engineering.

The π-conjugated core is the key to SAMs' hole extraction efficiency, as it delocalizes lone-pair electrons, enhancing electron density [12, 13]. Symmetric planar π-conjugated cores, such as those in 4PACz and Me-4PACz [1], enable ordered intermolecular arrangements and crystalline structures, which facilitate carrier extraction and transport. However, planar SAMs with low steric hindrance tend to aggregate in solution and film states, reducing solubility and causing uneven deposition [14]. This leads to serious interfacial losses and poor perovskite crystallization. Moreover, crystalline SAMs are prone to phase transformations or structural collapse under thermal or mechanical stress, disrupting carrier transport [15].

To mitigate aggregation and improve film uniformity, non-coplanar conjugated core structures have been developed. SAMs like 4PADCB and MeO-4PADCB introduce additional benzene rings to increase steric hindrance, while 4PA-Spiro and SAX use spiro-type designs to disrupt planarity [11, 15–17]. Similarly, Ph-4PACz and 4PADPhAC incorporate rotatable benzene groups to increase steric hindrance [18, 19]. These strategies inhibit aggregation and enhance SAM uniformity, with amorphous SAM films demonstrating superior stability under external stimuli. However, symmetric and non-coplanar SAMs often exhibit weak intermolecular forces due to small molecular dipoles and excessive steric hindrance, leading to sparse molecular arrangements, low interface coverage, and weak carrier transport [12]. This manifests as reduced open-circuit voltage (Voc) and fill factor (FF). Table S1 presents some recent work on the structural design of SAM molecules. Planar asymmetric molecular design strategies (e.g., DMICPA) have proven effective in suppressing molecular aggregation; however, the achievable dipole moments remain moderate (typically <2.5 D). Striking a balance between a dense non-aggregated arrangement and a substantially enhanced dipole moment remains a critical challenge [20, 38].

In this work, we develop a non-planar and asymmetric hole-selective molecular contact (named as ZF6), featuring an asymmetric dimethylindenocarbazole group for a large dipole moment and rotatable carbazole groups for a 3D non-planar structure. These properties regulate intermolecular interactions, promoting dense but non-aggregated arrangements. Compared to conventional SAM 4PACz, ZF6 exhibits tighter binding, an amorphous phase nature, and uniform coverage on transparent conductive substrate, reducing interfacial non-radiative recombination and enhancing carrier transport. Additionally, ZF6's amorphous structure ensures superior stability under external stimuli. Consequently, ZF6-based double-junction and triple-junction perovskite/silicon TSCs achieved certified steady-statePCEs of 32.81% and 30.42%, respectively. Moreover, ZF6-based TSCs demonstrate significantly improved operational stability, retaining over 97% and 87% of their initial efficiency after 1000 h of maximum power point tracking under continuous one-sun illumination at 45°C and 85°C, respectively. This work highlights the potential of asymmetric and non-planar SAMs to address key challenges in perovskite/silicon TSCs, paving the way for high-performance, stable, and commercially viable tandem photovoltaics.

2 | Results and Discussion

2.1 | Molecular Design and Self-Assembly Mechanisms

By modifying the classic carbazole-based SAM 4PACz, we designed and synthesized a stereoscopic asymmetric π-extended SAM, namely (4-(2-(9H-carbazol-9-yl)-7,7-dimethylindeno[2,1-b] carbazol-5(7H)-yl) butyl) phosphonic acid (ZF6). The molecular structures of 4PACz and ZF6 are depicted in Figure 1a. ZF6 incorporates an asymmetric π-conjugated dimethylindenocarbazole group and a rotatable carbazole moiety, conferring a large dipole moment and steric hindrance. These features promote a dense yet non-aggregated molecular arrangement, enhancing SAM adsorption on indium tin oxide (ITO; Figure 1b), which improves interface coverage and stability. The synthetic routes and nuclear magnetic resonance (NMR) spectroscopy profiles of ZF6 are detailed in Scheme S1 and Figures S1–S8.

Density functional theory (DFT) calculations reveal that the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) energy levels of ZF6 are -5.10 and -1.02 eV, respectively, compared with -5.35 and -0.67 eV for 4PACz (Figure S9). The absorption spectra (Figure S10) indicate optical bandgaps of 3.48 eV for 4PACZ and 3.25 eV for ZF6. Electrostatic potential (ESP) calculations (Figure S11) further show a higher charge density over the π-conjugated core of ZF6 than that of 4PACz, particularly on the singly linked carbazole. This enhanced charge delocalization benefits carrier transport and strengthens interactions with perovskite materials [21]. ZF6 extends the symmetric π-conjugated core of 4PACz by incorporating a dimethylindenocarbazole unit and introduces a single-bonded carbazole, thereby increasing molecular asymmetry. Consequently, the dipole moment of ZF6 is approximately doubled relative to 4PACz (3.93 D vs.1.89 D). The larger dipole originates from the asymmetric molecular architecture of ZF6, enabling stronger dipole-driven coupling at the perovskite interface, consistent with binding energy calculations between the SAM terminal groups and the perovskite surface (Figure 1b; Figure S12). We further evaluated the coordination interaction between SAMs and ITO using X-ray photoelectron spectroscopy (XPS). Compared to ITO/4PACz, the characteristic In 3d signal of ITO/ZF6 exhibits a 0.1 eV shift (Figure S13a), indicating stronger interaction between ZF6 and ITO. This was corroborated by binding energy calculations (-7.807 eV for ZF6 vs. -7.616 eV for 4PACz, Figure 1b). In both cases, the phosphonic acid anchoring group coordinates to surface In atoms through P–OH functionalities, effectively removing surface hydroxyl (–OH) species. Additionally, the expanded π-conjugated structure of ZF6 enhances the hydrophilicity of the modified ITO interface (Figure S14), facilitating enhanced perovskite crystallization (as discussed in later sections).

To elucidate the self-assembly mechanisms of these SAMs, we first analyzed intermolecular interactions using dimer models (Figure S15). The dimerization energies of 4PACz are significantly

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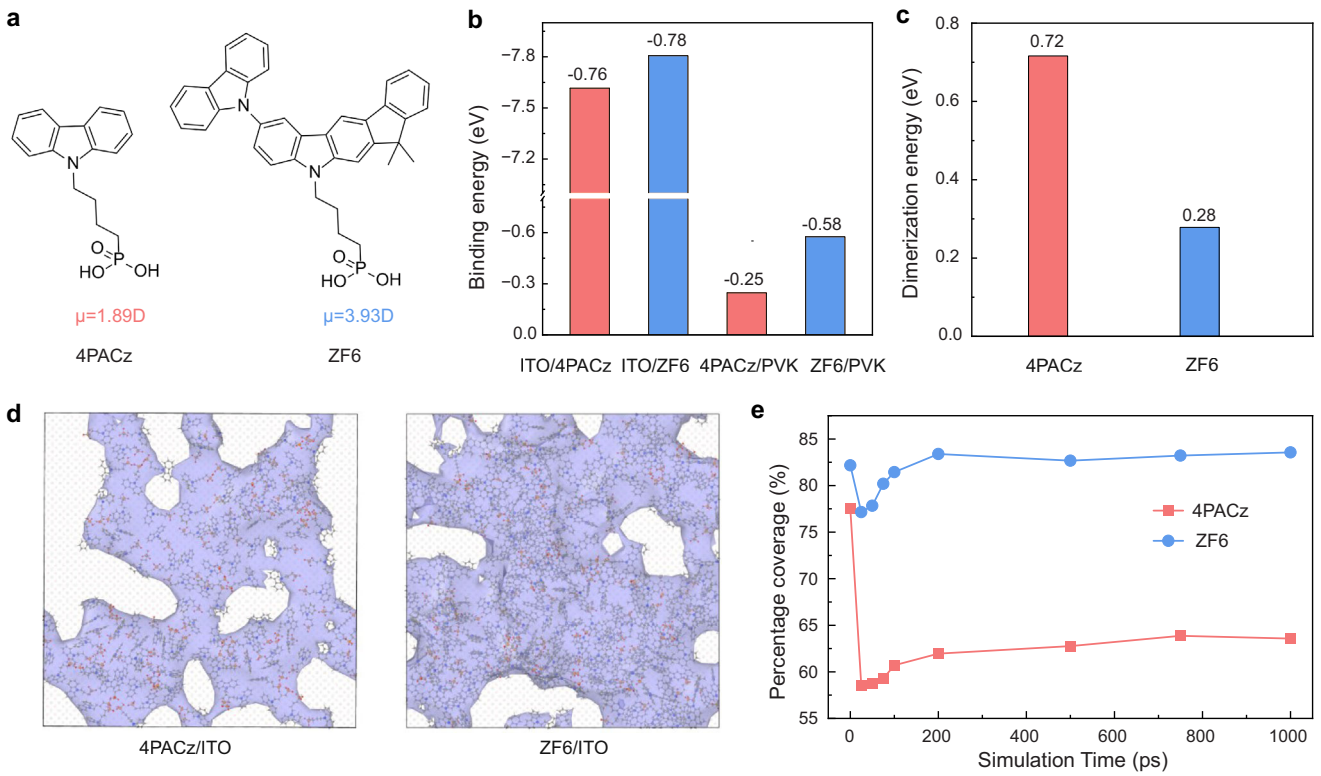


FIGURE 1 | Molecular design and self-assembly mechanisms of hole-selective layers (HSLs). (a) Molecular structures of 4PACz and ZF6 molecules. (b) Calculated binding energies for $FAPbI_3$ (100) and ITO surfaces adsorbed with 4PACz and ZF6 molecules, determined using DFT calculations at the GGA/PBE+vdW level. (c) Calculated dimerization energies of 4PACz and ZF6, highlighting intermolecular interactions. (d) Schematic representation of surface coverage for 4PACz and ZF6 on ITO substrates after 1000 ps molecular dynamics (MD) simulations. (e) Time-dependent surface coverage of 4PACz and ZF6 on ITO substrates, illustrating the evolution of molecular arrangement during the MD simulations.

larger than those of ZF6 (Figure 1c), indicating a stronger intermolecular association for 4PACz. The increased steric hindrance in ZF6 suppresses π–π stacking, thereby reducing aggregation while enhancing solubility and solution processability. We then investigated the self-assembly behaviors of 4PACz and ZF6 on ITO substrates by using molecular dynamics (MD) simulations. The adsorption process proceeds through three stages: physical adsorption, chemical adsorption, and the formation of a densely packed configuration [9, 22]. As illustrated in Figure 1d, 4PACz exhibits an intrinsic tendency to form clustered aggregates, whereas ZF6 is more uniformly distributed on the surface. Time-resolved coverage analysis (Figure 1e) reveals a pronounced decrease in surface coverage for 4PACz compared to ZF6. This reduction occurs predominantly within the first 100 ps (Figure S16). These simulation results align with the experimental observations (Figure S13b,c), where reduced molecular coverage at the microscopic level corresponds to lower macroscopic surface density for 4PACz.

2.2 | Amorphous Structure and Enhanced Thermal Stability

We investigated the crystallization characteristics of these SAMs by using X-ray diffraction (XRD) analysis. As shown in Figure 2a, 4PACz films exhibit distinct diffraction peaks, while ZF6 shows no obvious peaks, confirming its amorphous phase nature. This observation is further supported by grazing-incidence wide-angle X-ray scattering (GIWAXS), where 4PACz displays discrete Bragg spots indicative of crystallinity and molecular orientation, whereas ZF6 remains fully amorphous (Figure 2b). Polarization-dependent Raman spectroscopy provides additional insights (Figure 2c). For 4PACz, the Raman peak intensity at ~1,600 cm^{-1} (aromatic C═C stretching) varies significantly with polarization angle, reflecting structural anisotropy and crystalline order. In contrast, the corresponding peak in ZF6 shows no polarization dependence, indicating structural isotropy and its amorphous nature. The disordered arrangement of ZF6 enables the formation of uniformly stacked SAM layers with high coverage, addressing the critical challenges of interface uniformity and aggregation in perovskite solar cells.

The amorphous phase of ZF6 also enhances the thermal stability of HSLs. Ordered SAMs, such as 4PACz, tend to aggregate under external stimuli, leading to interface failure, while disordered SAMs maintain their amorphous phase, ensuring stability [7, 15]. We evaluated the thermal stability of ZF6 and 4PACz films on ITO using atomic force microscopy (AFM) before and after thermal aging (Figure 2d). After aging, 4PACz films exhibit obvious aggregates with the root mean square roughness (Rq) increasing from 2.80 to 9.85 nm, intensifying perovskite-ITO contact and interface recombination losses. In contrast, ZF6 films remain smooth and uniform morphology, with R_q increasing only slightly from 2.51 to 2.53 nm, demonstrating superior thermal stability. Kelvin probe force microscopy (KPFM) measurements of fresh SAM-modified ITO substrates revealed distinct surface

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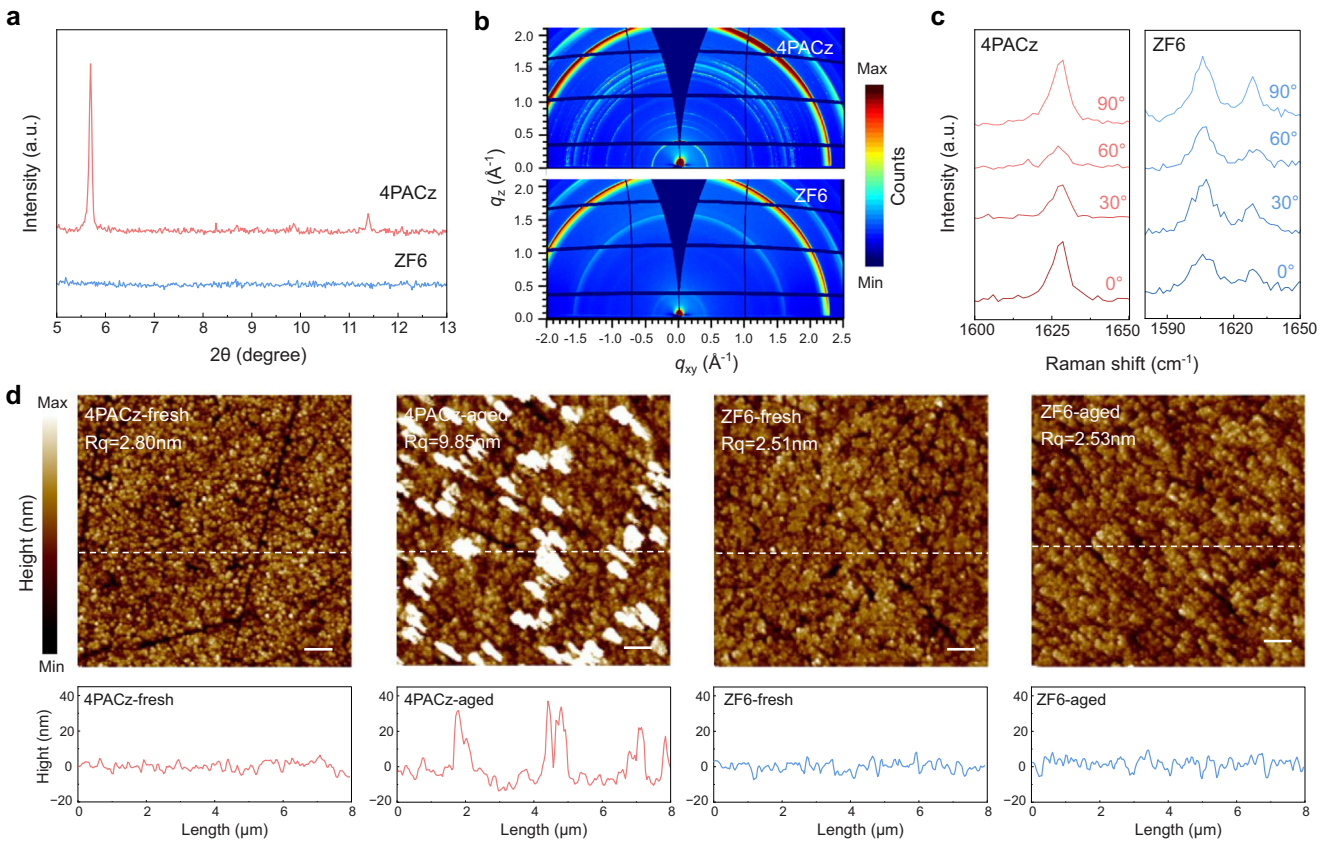


FIGURE 2 | Crystallization and thermal stability of HSLs. (a) XRD patterns, (b) 2D GIWAXS images, and (c) Raman spectra of 4PACz and ZF6 films on ITO substrate. (d) AFM images and the corresponding surface roughness of 4PACz and ZF6 films on ITO before and after thermal aging at 85°C for 400 h. The scale bar is 800 nm.

potential distributions (Figure S17). The 4PACz-modified ITO exhibits initial inhomogeneity, whereas ZF6 displayed uniform potential mapping. This suggests spontaneous self-aggregation of 4PACz during deposition, a tendency suppressed by ZF6. Under accelerated thermal aging (100°C, 600 h), post-aging AFM/KPFM characterization (Figure S18) showed persistent aggregation in 4PACz but maintained morphological homogeneity in ZF6. Notably, 4PACz aggregation sites exhibited amplified surface potential fluctuations. These observations indicate that thermal aging exacerbates surface potential heterogeneity through SAM desorption and aggregation, potentially increasing interfacial defects and impeding carrier transport. The uneven interface and insufficient coverage caused by SAM aggregation have long hindered the interfacial carrier dynamics and, consequently, the photovoltaic performance of perovskite solar cells [2, 6, 11]. This issue stems from the small steric hindrance of planar SAMs and the excessive intermolecular forces induced by π–π conjugation, which promote molecular aggregation and crystallization [20]. In contrast, ZF6 features an asymmetric π-conjugated extended structure and rotatable carbazole groups, providing significant steric hindrance that inhibits aggregation and favors disordered arrangements. Conversely, 4PACz, with its planar symmetric structure and low steric hindrance, tends to form crystalline domains.

To further elucidate the structural influence of ZF6 on SAM desorption behavior, we performed XPS analysis on both fresh and thermally aged (85°C, 400 h) SAM-functionalized ITO substrates after DMF rinsing. The SAM coverage factor, derived from P*2p*/In*3d* peak area ratios (Figure S19 and Table S2), revealed a 20.48% decrease for fresh 4PACz-modified ITO post-rinsing. This coverage loss intensified significantly in aged 4PACz samples. Conversely, ZF6-modified ITO exhibited minimal coverage reduction when fresh and maintained high coverage after aging. These results demonstrate that ZF6's asymmetric non-coplanar conjugated structure effectively suppresses SAM self-aggregation and desorption. Further analysis of perovskite buried interfaces aged at 85°C for 170 h (Figure S20) showed PbI_2 formation at grain boundaries in 4PACz-based devices, indicating interfacial decomposition. ZF6-based perovskites retained structural integrity, suggesting that ZF6's inhibition of thermal desorption prevents direct perovskite/ITO contact, thereby delaying decomposition.

2.3 | Improved Carrier Dynamics

We subsequently examined the carrier dynamics of ZF6 as an HSL in perovskite solar cells. The uniform and dense disordered stacking of ZF6 provides a smooth interface with fewer defects, promoting the growth of high-quality perovskite films. GIWAXS (Figure S21) and XRD (Figure S22) characterizations reveal that ZF6-based perovskite films exhibit sharper diffraction peaks and stronger scattering signals, indicating higher crystallinity compared to 4PACz-based counterparts. In situ photoluminescence (PL) spectra monitoring during perovskite spin-coating reveals substantially stronger emission intensity upon

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anti-solvent addition for ZF6-modified ITO substrates compared to 4PACz-modified counterparts (Figure S23). This enhanced emission indicates superior crystal nucleation efficiency on ZF6-functionalized surfaces, suggesting a mechanistic basis for improved perovskite film formation. AFM images (Figure S24) show that ZF6-based perovskite films have a smoother surface (Rq = 11.3 nm) than 4PACz-based perovskite films (Rq = 12.7 nm), which is attributed to the lower residual stress in the perovskite thin films fabricated based on ZF6 (Figure S25). To further understand the impact of SAMs on perovskite growth, we analyzed SEM images of the buried interface and statistically evaluated crystal grain distribution as illustrated in Figure S26. ZF6-based perovskite films exhibit smoother grain surfaces, attributed to the stronger bonding between ZF6 and the perovskite layer, driven by ZF6's larger dipole moment and superior surface coverage. Statistical analysis of grain distribution confirms that ZF6-based films have larger grain sizes and better crystallinity, facilitating carrier transport and reducing non-radiative recombination. These findings are further supported by PL mappings (Figure 3a), where ZF6-based perovskite films exhibit higher PL intensity and more uniform distribution than 4PACz-based films, highlighting their superior optoelectronic properties.

The π-conjugated structure can be tailored to align the energy level of the HSLs with those of the perovskite. To investigate this, we performed ultraviolet photoelectron spectroscopy (UPS) measurements on ITO modified with different SAMs (Figure S27). The ITO/ZF6 interface exhibits a smaller work function, consistent with KPFM results (Figure S17). Energy level alignment diagrams derived from SAM optical bandgaps and UPS measurements of 1.53, 1.68, and 1.95 eV bandgap perovskites (Figure S29) reveal that ZF6-modified ITO exhibits a HOMO level closer to the Fermi level than 4PACz. This configuration indicates enhanced p-type character at the interface, facilitating improved hole transport. Moreover, ZF6's HOMO demonstrates superior alignment with the valence band maxima across all three perovskite compositions, enabling more efficient hole extraction and reduced voltage losses. This is further supported by time-resolved photoluminescence (TRPL) measurements (Figure S30 and Table S3). The differential lifetime spectrum shows that ZF6-based perovskite transitions faster from the fast decay stage (dominated by charge transfer) to the slow decay stage (dominated by carrier recombination) [23], highlighting faster charge transfer at the ZF6/perovskite interface (Figure 3b). Transient absorption (TA) measurements on half devices with ITO/SAM/perovskite structure provide further insights into charge dynamics at the buried interface. The ground state bleaching (GSB) signal at ~720 nm decays faster in ZF6-based 1.68 eV bandgap perovskite films, indicating more efficient charge separation and transport (Figure S31). TA kinetics at the 720 nm peak, fitted with a two-exponential model, show that ZF6-based samples have a shorter effective lifetime (2.20 ns vs. 2.25 ns for 4PACz, Figure 3c; Table S4), confirming enhanced hole extraction and transport at the ZF6/perovskite interface [24]. This finding is further corroborated by TA spectra of 1.53 and 1.95 eV bandgap perovskites (Figures S31 and S32 and Table S4), demonstrating ZF6's universal efficacy in enhancing carrier transport across photovoltaic compositions.

We next analyzed the interfacial charge redistribution at the 4PACz/ITO and ZF6/ITO interfaces using charge density difference calculations. As shown in Figure 3d, both systems exhibit clear charge transfer, indicative of strong electronic coupling between the SAM and the ITO substrate. Compared with 4PACz/ITO, ZF6/ITO shows a slightly less sharpened charge-density profile (Figure 3e), consistent with reduced interfacial charge accumulation [7].

We further evaluated the photovoltaic performance of ZF6 by fabricating the single-junction PSCs with a p-i-n device structure: glass /ITO /HSL /perovskite /PCBM /zirconium acetylacetonate ($Zr(acac)_4$) /Ag. The ZF6-based 1.68 eV PSC delivered a PCE of 23.13%, with a short circuit current density (J_{SC}) of 21.30 mA cm^{-2}, a V_{OC} of 1.262 V, and an FF of 86.05%, outperforming the 4PACz-based device (PCE = 21.85%, J_{SC} of 20.71 mA cm^{-2}, V_{OC} of 1.260 V, FF of 83.60%), as shown in Figure 3f, Figure S33 and Table S5. ZF6 also demonstrated effectiveness as an HSL for 1.53 and 1.95 eV PSCs (Figure 3f; Figures S34 and S35), achieving reverse-scan PCEs of 26.79% and 13.77%, respectively, surpassing the 4PACz-based device (25.89%, 12.25%, respectively). ZF6-based devices also exhibit lower current hysteresis compared to 4PACz-based counterparts (Figure S36). Steady-state power output confirmed the reliability of the three bandgap devices, with stabilized PCEs of 25.99%, 21.99%, and 13.56% for 1.53, 1.68, and 1.95 eV, respectively (Figure S37). To assess the performance enhancement enabled by ZF6, intensity-dependent steady-state PL measurements were carried out on a half-device configuration: ITO/SAM/perovskite. Additionally, pseudo *J–V* curves were measured, and the corresponding implied open-circuit voltage (iV_{OC}) and pseudo-FF values of the half-devices were analyzed and summarized (Table S6). As shown in Figure 3g, the quasi-Fermi level splitting (QFLS) of the ITO/ZF6/perovskite stack was higher than that of the ITO/4PACz/perovskite stack, with iV_{OC} values of 1.180 V versus 1.174 V for the 1.53 eV perovskite films, 1.302 V versus 1.290 V for 1.68 eV perovskite films, and 1.431 V versus 1.424 V for the 1.95 eV perovskite films, respectively. This indicated that the ITO/perovskite interface modified with ZF6 exhibited lower voltage loss. We also analyzed the FF loss of the three bandgap devices based on the pseudo-FF results (Figure 3h). Among the devices, ZF6-based devices showed significantly reduced transport losses compared to 4PACz-based devices, likely due to the faster carrier extraction, as supported by TA measurements and charge density difference calculations. Collectively, the enhanced carrier extraction and decreased interface trap state density in ZF6 single-junction PSCs suppressed the non-radiative recombination losses, which will also be beneficial for the development of perovskite-based TSCs.

2.4 | Photovoltaic Performance of Monolithic Perovskite/Silicon TSCs

Monolithic double-junction perovskite/silicon TSCs were fabricated using silicon heterojunction bottom cells and 1.68 eV WBG perovskite top cells, with ZF6 or 4PACz as HSL (as detailed in Methods). The cross-sectional SEM image of the double-junction TSC is shown in Figure 4a. To evaluate the photovoltaic performance of these TSCs, more than ten independent devices were fabricated. As illustrated in Figure S38 and Table S7, the ZF6-based TSCs exhibited a higher average PCE of 33.83 ± 0.17% compared to the 4PACz-based TSCs (32.87 ± 0.46%), attributed to improved FF and V_{OC} values. The champion ZF6-based TSC

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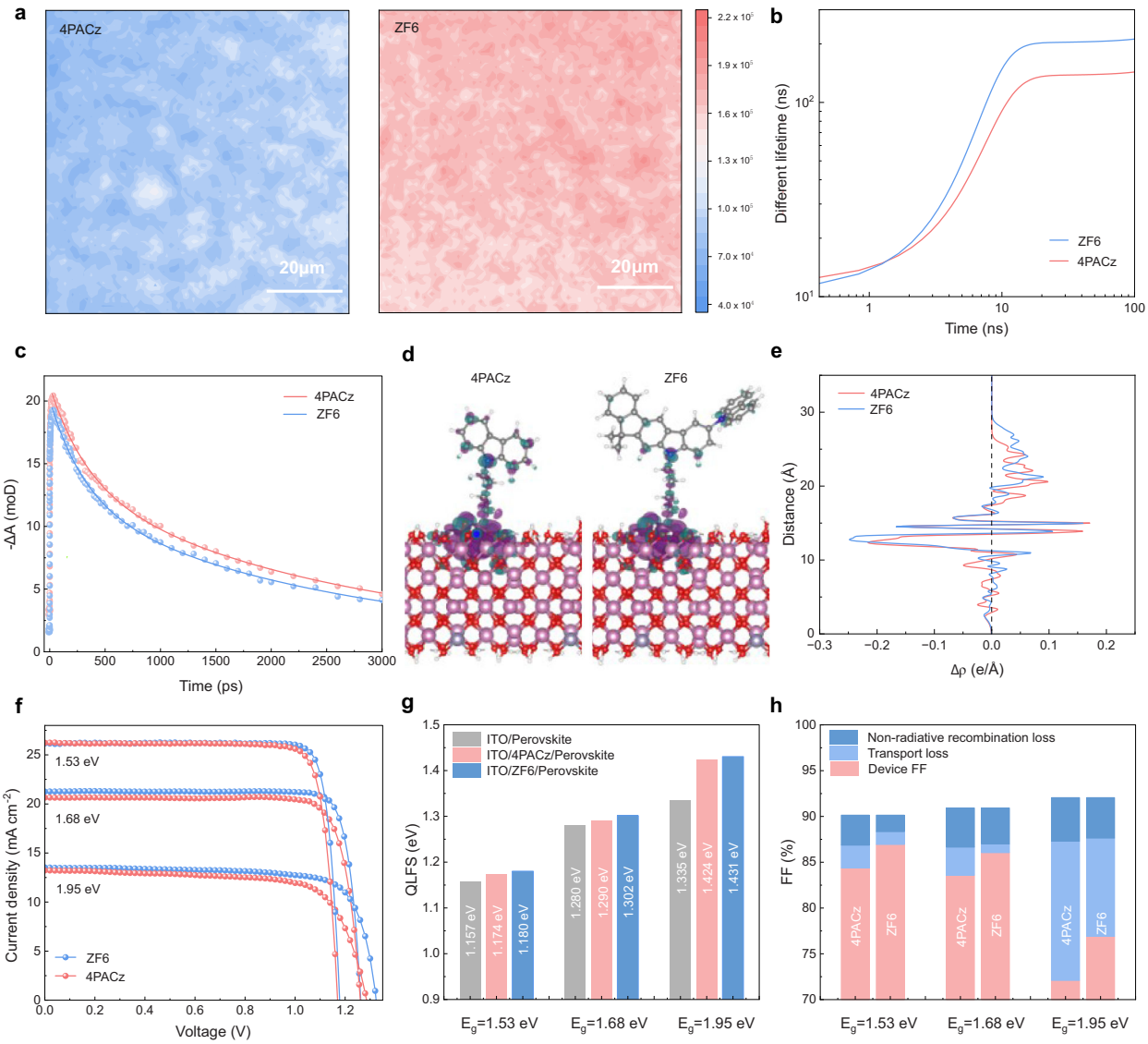


FIGURE 3 | Carrier dynamics. (a) PL mappings and (b) TRPL of 1.68 eV perovskite films deposited on 4PACz/ITO and ZF6/ITO substrates. (c) Transient absorption (TA) kinetics of perovskite films deposited on 4PACz/ITO and ZF6/ITO substrates at 740 nm peak. (d) Charge density differences (isovalue = 0.0015 e/Å [3]) and (e) the corresponding 1D planar charge density differences ($\Delta\rho$) along the z-axis for 4PACz/ITO and ZF6/ITO interfaces. (f) J–V curves of best-performing ZF6- and 4PACz-based single-junction PSCs with bandgaps of 1.53, 1.68, and 1.95 eV. (g) QFLS and (h) FF loss analysis in the case of three bandgaps of perovskite with different HSLs.

achieved a reserve-scan PCE of 34.02% (independently certified at 32.81% by NREL, Figure S39), with a V_{OC} of 1.999 V, a J_{SC} of 20.31 mA cm^{-2}, and an FF of 83.78%. This device exhibits negligible hysteresis (Figure 4b) and significantly outperforms 4PACz-based counterparts, demonstrating superior photovoltaic performance. The external quantum efficiency (EQE) spectra of the ZF6-based double-junction perovskite/silicon TSCs is presented in Figure 4c. The integrated photogenerated current densities for the perovskite top sub-cell and silicon bottom sub-cell were 20.96 and 20.30 mA cm^{-2}, respectively, consistent with J_{SC} derived from the J–V measurements.

Additionally, triple-junction perovskite/perovskite/silicon TSCs were fabricated using ZF6 as the HSLs, incorporating an ultra-WBG perovskite top cell, a middle-bandgap (MBG) perovskite middle cell, and a silicon heterojunction bottom cell (as detailed in Methods). The device configuration of the triple-junction TSC is illustrated in Figure 4d. ZF6-based triple-junction TSCs achieved an average PCE of 29.90 ± 0.32% (V_{OC} = 3.15 ± 0.01 V, J_{SC} = 11.67 ± 0.14 mA cm^{-2}, FF = 81.27 ± 1.14%) as documented in Figure S40 and Table S8. This represents a significant enhancement over 4PACz-based devices, underscoring ZF6's materials versatility in multi-junction architectures. The champion ZF6-based triple-junction TSC was sent to the National Institute of Metrology (NIM), an accredited independent PV calibration laboratory, for certification, where it achieved a certified reverse-scan PCE of 30.26% and steady-state PCE of 30.42% (Figure 4e; Figure S41). The EQE spectra of individual sub-cells (Figure 4f) reveal balanced photocurrent generation, with values of 12.38 mA cm^{-2} (WBG perovskite), 11.91 mA cm^{-2} (MBG perovskite), and 16.48 mA cm^{-2} (silicon bottom cell). To the best of our knowledge, this result represents the highest-performing perovskite-based

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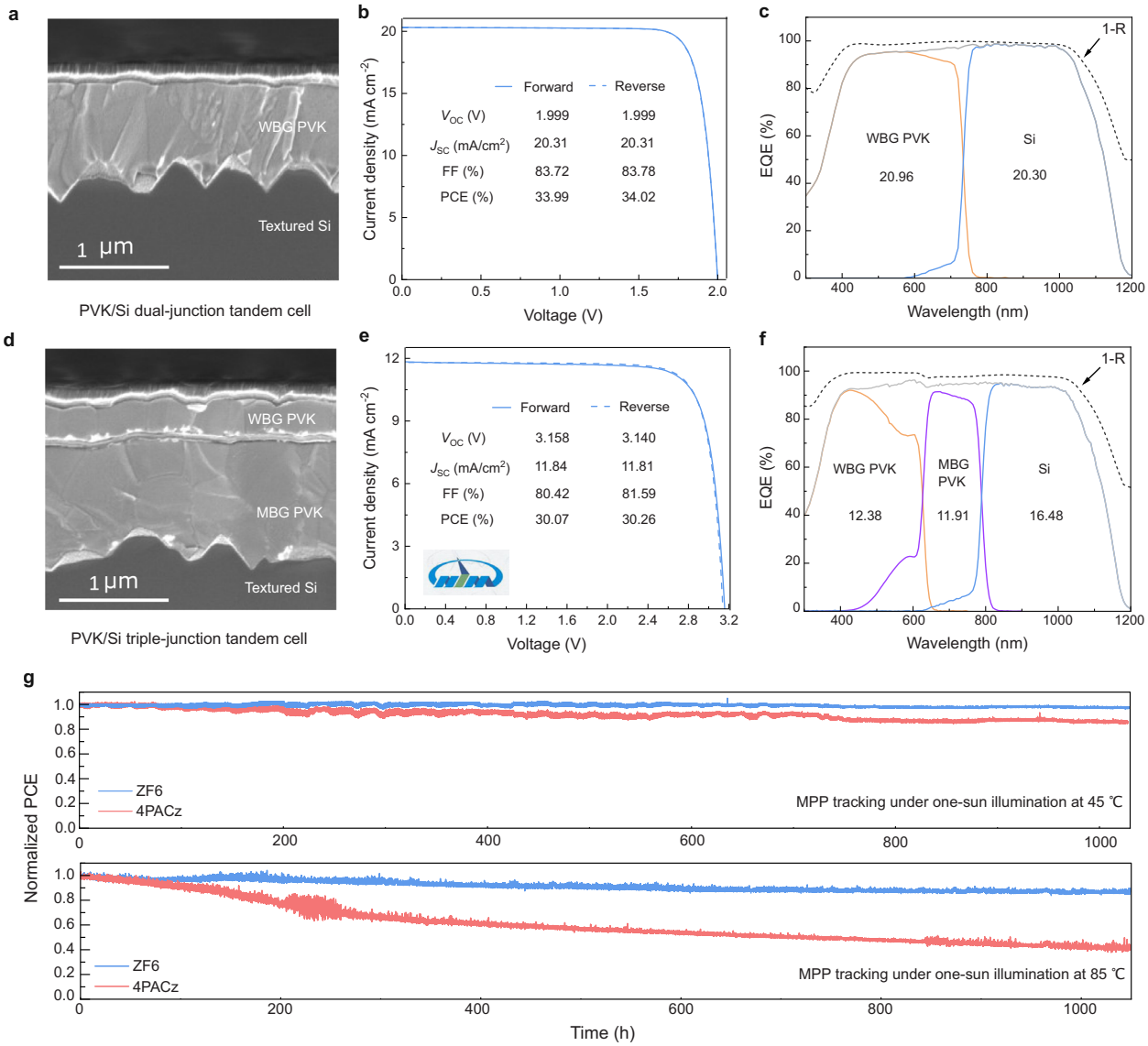


FIGURE 4 | Photovoltaic performance of monolithic perovskite/silicon TSCs. (a) Cross-sectional SEM image, (b) *J–V* curves, and (c) EQE spectra of the best-performing ZF6-based double-junction perovskite/silicon TSCs. (d) Cross-sectional SEM image, (e) certified *J–V* curves, and (f) EQE spectra of the best-performing ZF6-based triple-junction perovskite/silicon TSCs. (g) Maximum power point (MPP) tracking of the encapsulated double-junction TSCs measured at 45°C and 85°C under continuous one-sun illumination in an ambient environment (relative humidity: 60 ± 10%).

triple-junction TSCs reported to date [4, 25], as summarized in Table S9.

We also investigated the long-term operational stability of TSCs under accelerated-aging conditions following the International Summit on Organic Photovoltaic Stability (ISOS) protocols. Encapsulated double-junction perovskite/silicon TSCs based on ZF6 and 4PACz were aged at 45°C and 85°C under continuous one-sun illumination to evaluate their photo-thermal stability. As depicted in Figure 4g, the ZF6-based TSC retained over 97% of its initial efficiency after 1020 h at 45°C, significantly outperforming the 4PACz-based TSC, which retained approximately 86%. Remarkably, under 85°C thermal stress, ZF6-based TSC retained over 87% of initial PCE after 1050 h, starkly outperforming 4PACz-based counterparts (~42% retention). Comparative analysis of pre-/post-aging *J–V* characteristics and *V*oc evolution (Figures S42 and S43) confirms ZF6's efficacy in suppressing voltage degradation. This enhanced stability is attributed to ZF6's molecular structure, which impedes self-aggregation and interfacial desorption mechanisms. These results underscore the superior stability of ZF6-based TSCs under high temperatures and prolonged illumination. The findings highlight the effectiveness of the molecular design featuring an asymmetric π-extended structure with an amorphous phase, demonstrating its potential as a robust approach for developing efficient SAMs and advancing perovskite-based TSC technology.

3 | Conclusion

In conclusion, we have successfully developed a novel self-assembled molecule (named as ZF6), by integrating asymmetric π system expansion and amorphous strategies. This innovative approach enables the formation of dense, uniform, and

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disordered SAM arrangements on ITO substrates, significantly enhancing interfacial carrier transport and stability. Comprehensive characterizations reveal that ZF6 exhibits amorphous film morphology, stronger substrate anchoring, optimized energy-level alignment with WBG perovskites, and superior coverage and uniformity. These attributes facilitate the growth of high-quality WBG perovskite films with reduced defects and improved carrier dynamics. As a result, ZF6-based double-junction perovskite/silicon TSCs achieved a certified steady-state PCE of 32.81% and exhibited superior stability under various aging conditions. Additionally, ZF6-based triple-junction perovskite/silicon TSCs reached a record-certified steady-state PCE of 30.42%. This study establishes a rational design paradigm for high-performance SAMs, paving the way for advancing the efficiency and stability of perovskite-based TSCs toward commercialization.

4 | Methods

4.1 | Materials

Lead iodide (PbI_2, 99.999%), lead bromide ($PbBr_2$, 98%) and lead chloride ($PbCl_2$, 99.99%) were purchased from Advanced Election Technology Co., Ltd. Dimethylammonium iodide (DMAI), cesium iodide (CsI), formamidinium iodide (FAI), ethane-1,2-diammonium iodide (ethylenediammonium diiodide, $EDAI_2$), and propane-1,3-diammonium diiodide (PDADI) were purchased from Greatcell Solar. 4PACz and Zirconium(IV) acetylacetonate ($Zr(acac)_4$) were purchased from Xi'an Polymer Light Technology. [6,6]-phenyl-C61-butyric acid methyl ester ($PC_{61}BM$, > 99.5%) were purchased from Nano-C. All solvents were purchased from Sigma–Aldrich. The above materials were used as received without further purification.

4.2 | Device Fabrication

4.2.1 | Preparation of Perovskite Precursor

The 1.68 eV perovskite precursor solution was prepared by dissolving 0.10 M DMAI, 0.30 M CsI, 0.60 M FAI, 0.02 M $PbCl_2$, 0.20 M $PbBr_2$, 0.81 M PbI_2 in 1 mL dimethylformamide (DMF)/dimethyl sulfoxide (DMSO) mixed solvent (v/v 4:1). The 1.53 eV perovskite precursor was prepared by dissolving 1.54 M PbI_2, 0.075 M CsI, 1.425 M FAI and 0.02 M $PbCl_2$ dissolved in a in 1 mL DMF/DMSO mixed solvent (v/v 4:1). The 1.95 eV perovskite precursor solution was prepared by dissolving 0.60 M PbI_2, 0.20 M CsI, 0.80 M FAI, 0.90 M $PbBr_2$, 0.02 M $PbCl_2$ was dissolved in 1 mL DMF/DMSO mixed solvent (v/v 4:1). All the perovskite precursor solutions were shake for 1–2 h until completely dissolved.

4.2.2 | Fabrication of Single-Junction Perovskite Solar Cells

The pre-patterned indium tin oxide (ITO) substrates (Advanced Election Technology Co., Ltd, 15 Ω sq^{-1}) were cleaned and treated with UV–ozone for 40 min for the next step. For SAM deposition, 4PACz and ZF6 with a concentration of 0.3 mg/mL in ethanol (EtOH) were spin-coated at 4000 rpm for 30 s on the cleaned ITO substrates and annealed at 100°C for 10 min. The perovskite films were deposited in the nitrogen (N_2)-filled glove box with controlled H_2O and O_2 levels below 0.1 ppm, and the temperature was monitored to be 23°C. 50 μL of perovskite precursor was dropped onto the HSL substrate. The substrate was spun at 2000 rpm for 15 s with an acceleration of 1000 rpm/s at first, and then at 4000 rpm for 45 s with an acceleration of 2000 rpm/s. The nitrogen gun was vertically positioned 5 cm above the top of the substrates, and the gas flow started after 30 s of the spin and a duration of 20 s when the gas flow pressure was 30 psi. The 1 mg/mL of EDAI solution in isopropyl alcohol (IPA) and DMF solution (v/v 150:1) was spin-coated onto the perovskite surface at 4000 rpm for 30 s and annealed at 70°C for 5 min. The 23 mg/mL of $PC_{61}BM$ solution in chlorobenzene (CB) was spin-coated at 2500 rpm for 40 s and then annealed at 70°C for 10 min. Then, 2 mg $Zr(acac)_4$ was added into 1 mL IPA and was spin-coated on $PC_{61}BM$ film at 5000 rpm for 30 s. Finally, a 100 nm silver (Ag) electrode with an active area of 0.1017 cm^2 was deposited by thermal evaporation. All the solutions were used after filtration.

4.2.3 | Fabrication of Silicon Bottom Cells

The silicon bottom cells were fabricated using n-type Czochralski (CZ) silicon wafers with a resistivity of 1–5 Ω/cm and a thickness of approximately 250–300 μm. Initially, a double-sided symmetrical texturing process was then performed using a potassium hydroxide (KOH) solution, resulting in random pyramids with a height of 500–1000 nm on both surfaces. Subsequently, the wafers underwent a standard RCA cleaning procedure and were then loaded into a plasma-enhanced chemical vapor deposition (PECVD) cluster for the passivation and doped layers. Hydrogenated intrinsic amorphous silicon (i-a-Si:H) layers, each with a thickness of about 10 nm, were deposited on both the front and rear surfaces. This was followed by the deposition of a 20-nm-thick n-type hydrogenated nanocrystalline silicon (n-nc-Si:H) layer and a 30-nm-thick p-nc-Si:H layer on the front and rear side of the wafer, respectively. Subsequently, a 150-nm-thick ITO layer was sputtered onto the rear side through a shadow mask defining an active area of 1.2 × 1.2 cm^2. Ag rear electrodes were then screen-printed onto the ITO layer, also through the same mask pattern. Finally, a 10-nm-thick ITO layer, serving as the tunnel recombination layer, was deposited on the front side. The completed silicon bottom cells were laser-cut into 2.5 × 2.5 cm^2 substrates for subsequent tandem cell fabrication.

4.2.4 | Fabrication of Double-Junction Perovskite/Silicon TSCs

The in-house-made n-i-n bottom wafer was treated with UV–ozone for 20 min. 0.6 mg/mL SAMs solution in ethanol (EtOH): DMF (v/v 3:1) was spin-coated at 3000 rpm for 30 s with an acceleration of 1000 rpm/s, and annealed at 100°C for 10 min. Then, 100 μL of the 1.68 eV perovskite precursor solution (1.7 M) was spin-coated onto the substrates at 600 rpm for 6 s with an acceleration of 200 rpm/s, followed by 2000 rpm for 50 s with an acceleration of 2000 rpm/s, and 6000 rpm for 15 s with an

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acceleration of 2000 rpm/s. During the spin-coating process, 300 µL of anisole was dropped onto the film at 59–60 s. The resulting perovskite film was immediately annealed at 100°C for 20 min. Subsequently, a 1 mg/mL of $EDAI_2$ solution in IPA was spin-coated onto the perovskite films at 5000 rpm for 30 s, followed by annealing at 100°C for 5 min. The completed substrates were transferred into a thermal evaporation chamber, where 9 nm of C_{60} was deposited at a rate of 0.05–0.1 Å s^{-1}. Subsequently, 21 nm SnO_x was grown in an atomic layer deposition (ALD) chamber at 80°C. After ALD process, 40 nm IZO was sputtered. Finally, 400 nm of Ag and 90 nm MgF_x was deposited in a thermal evaporation chamber. The final devices with a structure of silicon/ITO/SAM/perovskite/C_{60}/SnO_2/IZOAg/MgF_x had an active area of 1.21 cm^2. All the solutions were used after filtration.

4.2.5 | Fabrication of Triple-Junction Perovskite/Silicon TSCs

The in-house fabricated heterojunction solar cells with 8 nm ITO on n-type nano-crystallized silicon were first treated by UV–ozone for 15 min. Then, 0.6 mg/mL ZF6 solution in EtOH: DMF (v/v 3:1) was spin-coated on the bottom silicon solar cells at 3000 rpm for 30 s with an acceleration of 1000 rpm/s, and annealed at 100°C for 10 min. The middle perovskite layers deposited were onto the HSL substrates at 600 rpm for 6 s with an acceleration of 200 rpm/s, followed by 2000 rpm for 50 s with an acceleration of 2000 rpm/s, and 6000 rpm for 15 s with an acceleration of 2000 rpm/s. During the spin-coating process, 300 µL of anisole was dropped onto the film at 59–60 s. The resulting perovskite film was immediately annealed at 100°C for 20 min. Then, 20 nm C_{60} was deposited by thermal evaporation. After that, 30 nm compact SnO_x was grown at 80°C with an ALD system. After ALD process, 20 nm IZO was sputtered. After that, 0.6 mg/mL ZF6 solution in EtOH: DMF (v/v 3:1) was spin-coated at 4000 rpm for 15 s, and then annealed at 100°C for 10 min. The top wide bandgap perovskite active layers were deposited as the same method for middle perovskite layer. Then, 0.5 mg/mL PDADI in IPA was dynamic spun onto the perovskite and annealed at 100°C for 10 min. Then, 10 nm C_{60} was deposited by a thermal evaporation. After that, 20 nm SnO_x was grown at 80°C. After the ALD process, 40 nm IZO was sputtered. Finally, 400 nm Ag and 100 nm MgF_x were deposited by thermal evaporation. The final devices with a structure of silicon/ITO/ZF6/perovskite/C_{60}/-SnO_2/IZO/ZF6/perovskite/C_{60}/SnO_2/IZO/Ag/MgF_x had an active area of 1.21 cm^2. All the solutions were used after filtration.

4.2.6 | Encapsulation Process of Tandem Device

The perovskite-silicon tandem device was initially dried at 60°C for 5 min on a hotplate to remove moisture residuals. A layered structure was then assembled sequentially with glass cover, butyl rubber sealant, photovoltaic device, and top glass cover, where silver conductive tapes were pre-attached to electrode terminals for external connection. The stack was transferred into a vacuum laminator chamber, where degassing was conducted under reduced pressure (<100 Pa) for 5 min, followed by lamination at 100°C under constant pressure for 5 min. Finally, edge sealing was applied, and the encapsulated device was cooled to ambient temperature (25°C) to complete the process.

4.3 | Characterizations

4.3.1 | Characterizations of Solar Cell

The photovoltaic performance of perovskite solar cells was characterized by current density-voltage measurement (*J–V*) using an AM 1.5G simulator in the N_2-filled glove box with controlled H_2O and O_2 levels below 0.1 ppm. *J–V* information was collected by Keithley 2450. The intensity of the simulator was calibrated to one-sun (100 mW/cm^2) using an NREL-calibrated standard silicon solar cell simulator before testing. The device area was 0.1017 cm^2 (R = 0.18 cm). Unless otherwise stated, all perovskite single-junction devices were masked with an area of 0.0804 cm^2 (R = 0.16 cm) metal aperture to define the active area and to eliminate edge effects. The voltage step was 0.02 V, the scan rate was 20 mV/s in the range of −0.1–1.4 V or 1.4–−0.1 V. The devices were measured under a constant voltage near the maximum-power-point (MPP) of *J–V* scans and recording the continuous current output for the stabilized power output (SPO). For perovskite/silicon TSCs measurements, the voltage step is 0.02 V. The current–voltage was measured with a scan rate of 20 mV/s ranging from 2.5 or 3.5 to -0.1 V and then reversed again from -0.1 to 2.5 or 3.5 V in a nitrogen atmosphere. And the cells were masked with a black metal mask with an area of 1.05 cm^2. External quantum efficiency (EQE) was recorded on a solar cell quantum efficiency measurement system supported by Enli Technology. A calibrated Si diode with a known spectral response was used as a reference. For stability test, maximum-power-point (MPP) tracking of tandem cells was measured under white light-emitting diode with light intensity of 100 mW/cm^2 at 45°C or 85°C in a temperature-controlled chamber. The light intensity was calibrated by measuring the current of the silicon solar cell.

4.3.2 | Other Characterizations

Scanning electron microscopy (SEM) images of perovskites were measured by a field-emission scanning electron microscope at 5 kV (ZEISS Sigma 300HD). The X-Ray Diffraction (XRD) patterns were recorded on a Bruker D8 Advance diffractometer with Cu Kα (λ = 1.5418 Å) as the X-ray source. The Atomic Force Microscope (AFM) and Kelvin probe force microscope (KPFM) images of samples were measured by a Dimension Icon (Bruker). The probe model is scm-pit-v2, and the probe work function corrected by highly oriented pyrolytic graphite (HOPG) is 4.8 eV. The grazing incidence wide-angle X-ray scattering (GIWAXS) was evaluated at the BL17B1 beamline of the Shanghai Synchrotron Radiation Facility (SSRF) by X-ray with a wavelength of 1.24 Å. The X-ray beam size (~0.2 mm) is determined by the pinhole. GIWAXS patterns were obtained by a PILATUS detector with a distance of ~263 mm from the material, and the exposure time was 30 s. The FIT2D software was used to analyze the GIWAXS patterns, and the scattering vector q coordinate was used to display it. Ultraviolet Photoelectron Spectroscopy (UPS) and (X-ray photoelectron spectroscopy) XPS measurements were performed

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using an ultrahigh vacuum surface analysis system equipped with an ULVAC-PHI 5000 VersaProbe III spectrometer. UPS was performed with He I (21.22 eV) acting as the excitation source at an energy resolution of 50 meV. XPS used monochromatic Al Kα radiation (1486.6 eV). The Steady-state photoluminescence (SSPL) data was obtained from Fluorolog-QM produced by HORIBA. PL mapping and time-resolved PL (TRPL) of the perovskite films were measured using MAPS-2024-Mark1-FLIM produced by Waynelabs Instruments & Solutions. The TRPL spectra were acquired at 500kHZ with the laser wavelength of 512 nm. The ultraviolet–visible spectroscopy (UV–vis) absorption spectra of the samples were obtained by a UV–vis diffuse reflectance spectrophotometer (Varian Carry 500 spectrometer, USA). Quasi-Fermi-level splitting (QFLS) and *pseudo-JV* data were obtained from QFLS-mapper (Enli Technology). Femtosecond transient absorption (fs-TA) spectroscopy was carried out with a Yb:KGW laser (1030 nm, 10 W, 33 kHz, OPA 303 μJ, Light Conversion Ltd). Its fundamental output was split into two beams: one was sent to a non-collinear optical parametric amplifier (NOPA, ORPHEUS-N, Light Conversion Ltd) to produce a 440 nm pump pulse (this study's target wavelength), the other was focused on a YAG crystal to generate a white-light continuum as a probe pulse. The pump and probe pulses overlapped spatially on the sample at an angle <10°, and the transmitted probe was collected by a linear CCD array. Polarized Raman measurements were performed on a Renishaw (inVia Qontor) Raman system. ^{1}H NMR spectra of the samples were obtained by NMR spectrometer (400 MHz, Bruker).

4.4 | Computational Methods

4.4.1 | Molecular Dynamics (MD) Simulations

Classical molecular dynamics simulations of the 4PACz/ITO and ZF6/ITO systems were performed using the large-scale atomic/molecular massively parallel simulator (LAMMPS) [26]. Intermolecular parameters for both organic molecules were assigned using the OPLS-AA force field generated via the LigParGen server [27]. The phosphonic acid anchoring groups were described with specialised parameters [6] to capture interactions between the ITO surface and SAM molecules. The ITO substrate was simplified to a fixed In_2O_3 model to avoid surface instabilities [22, 28]. The slab dimensions were 103.0 Å × 103.0 Å × 20.6 Å, and the substrate interactions were described using a Buckingham potential forcefield [29]. All non-bonding interactions between the organic molecules and the substrate were simulated and modeled using Lennard–Jones (LJ) and Coulombic terms with geometric mixing rules, as implemented in the Universal Forcefield (UFF) [30], using cutoffs of 10.0 Å (van der Waals) and 8.0 Å (Coulombic interactions). Periodic boundary conditions were applied in the x and y directions. Long-range electrostatic interactions were treated with the particle-particle-particle-mesh (PPPM) method with a force accuracy of 1×10^{-4} kcal·mol^{-1}·Å^{-1}. Both systems contained 120 molecules, initialized with a minimum intermolecular separation of 3.5 Å using the Packmol package [31]. Surface coverage analysis was performed using the alpha-shape method and visualized using OVITO. All simulations were equilibrated in the Nosé–Hoover thermostat (NVT) using a velocity–Verlet algorithm at 300 K for 1.0 ns with a 1 fs timestep. Configurations were saved every 1000 fs for subsequent structural analysis and visualization.

4.4.2 | Density Functional Theory (DFT) Calculations

Geometry optimization and harmonic vibrational frequency calculations of SAM molecules (4PACz and ZF6) were performed at the B3LYP/def2-SVP level using the Gaussian 09 package (version C01) [26]. Dimerization energies of SAM molecules were calculated using high-level DLPNO-CCSD(T) as implemented in the ORCA package [32]. For the periodic interface calculations of SAM/$FAPbI_3$ (100) and SAM/ITO (100), the DFT calculations were performed using the projector-augmented wave (PAW) method as implemented in Vienna Ab initio Simulation Package (VASP, version 6.5.0) [33, 34]. The Perdew–Burke–Ernzerhof (PBE) exchange-correlation functional within the generalized gradient approximation (GGA) was employed, together with Grimme's D3 dispersion corrections [35, 36]. The plane-wave cutoff energy of 450 eV was used, and the 2 × 2 × 1 Monkhorst-Pack k-point was employed for Brillouin zone sampling. All atomic positions were fully relaxed with a convergence criterion of 10^{-5} eV for the total energy and 0.015 eV/Å for the residual forces per atom. The Bader charge analysis was performed to evaluate the charge transfer between SAM and ITO.

4.4.3 | Average Carrier Lifetime

The TRPL and TA results were fitted with a double-exponential model to obtain A_1, τ_1, A_2, and τ_2. The average carrier lifetime (τ_{ave}) was calculated according to the equation [37].

$$\tau_{ave} = \frac{\sum A_i \tau_i^2}{\sum A_i \tau_i}$$

Author Contributions

H.Z. was convinced the idea. Y.F.C. (Yifan) and D.R.J. fabricated and characterized most of the samples and devices. H.Z., J.F.G., and Y.J. supervised the project. X.P.Z. is responsible for the synthesis and characterization of the ZF6 molecule. A.L. and J.Y. performed the theoretical simulations. Y.G.Y. performed the GIWAXS measurements and analyzed the results. Z.J.W., M.L., Q.Q.W., and C.Y.X. conducted the PL, PLQY, and QFLS measurements. X.J.H. performed the AFM and KPFM measurements. Y.M. fabricated the silicon bottom cells. Z.L., R.X., R.H., J.X., H.J.L., G.T.Y., and D.N.Y. fabricated and characterized the double-junction TSCs. Y.X. fabricated and characterized the triple-junction TSCs. Y.F.C. (Yifan) and L.C.G. conducted the long-term operational stability tests. Y.F.C., C.Y.X., and H.Z. wrote the first draft of the manuscript. W.C., J.H.C., Y.F.C. (Yifeng), X.L.Z., and D.H.M. were involved in the data analysis and discussions and revised the manuscript. All authors discussed and finalized the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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